Strict path shadowing — cross-method comparison

Five nested-prefix bank sizes, one 1 000 000-path bank

`results/Heston/preprocessing_with_log_returns`
BASSERAS/benchmark

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How RMSE and MAE are aggregated

Fix a query path $q \in \{1, \dots, m\}$ ($m = 512$) and a horizon offset $u \in \{1, \dots, H\}$ ($H = 32$). The forecast is the mean over the $K = 256$ retrieved continuations, so the per-cell squared error and its horizon average are

$$se_{q,u} = \left(\tfrac{1}{K}\sum_{k=1}^{K}\hat{Y}_{q,u,k} - y_{q,u}\right)^2, \qquad se_q = \tfrac{1}{H}\sum_{u=1}^{H} se_{q,u}.$$

There are two ways to finish, and they are *not* equal. Taking the root last gives the textbook RMSE; taking it inside first gives the mean of per-path root-mean-square errors:

$$\underbrace{\text{RMSE} = \sqrt{\frac{1}{m}\sum_{q=1}^{m} se_q}}_{\text{root last — reported below}} \qquad \underbrace{\frac{1}{m}\sum_{q=1}^{m}\sqrt{se_q}}_{\text{root inside}}\,.$$

Because $\sqrt{\cdot}$ is concave, Jensen's inequality makes the root-inside form *always* the smaller of the two; it is a real statistic, but it is not an RMSE. **The `cum` and `step` rows below report the textbook (root-last) form.**

Why `rv` gets two rows. Horizon realised volatility is a *scalar* per query ($H = 1$), so the inner mean collapses and the two aggregations above reduce to two different standard norms of the same 512-vector of signed errors e_q:

$$\text{MAE} = \frac{1}{m}\sum_{q=1}^{m}|e_q|, \qquad \text{RMSE} = \sqrt{\frac{1}{m}\sum_{q=1}^{m} e_q^2}.$$

Both are listed. For CSDI at the 1 000 000-path bank they are **0.01373** and **0.01772** — a ratio of 1.29, against 1.06 for `step`. That ratio is a direct read-out of how much of the error sits in the tail: it equals 1 when every query errs by the same amount and grows with error dispersion. rv is the quantity where the generators separate, and it is also the quantity whose error distribution is the most skewed — which is exactly why quoting one norm and labelling it the other is not a cosmetic issue.

Reading the tables. Lower is better everywhere except coverage, whose target is the nominal level. *Oracle* is the Heston ground-truth ceiling (retrieval from the true data-generating process, size-matched to the bank); *RW* is the random-walk floor. The Winner column ranks the generators and forecasters only — never the oracle or the floor — and the two `width` rows are diagnostic, so they name no winner. That leaves **18 ranked rows** per table (6 ranked metrics × 3 quantities).

† The rv root-last RMSE row is bolded and its winner named in *italics*, but it is **not counted** in the 18-row total: ranking two norms of one error vector would double-count rv against `cum` and `step`.

Bank size $N = 4\,096$

Winners (of 18 ranked rows): TimeDiT (raw) 15, SBTS 2, CSDI 1.

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ **trajectory)**									
RMSE ↓	0.04919	**0.04890**	0.04906	0.04916	0.04921	0.05094	0.04924	0.05670	**TimeDiT (raw)**
CRPS ↓	0.02548	**0.02531**	0.02544	0.02544	0.02624	0.02722	0.02539	0.02946	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4571	**0.5105**	0.4764	0.4839	0.4050	0.4664	0.4875	0.4719	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8629	**0.8957**	0.8691	0.8805	0.7905	0.8177	0.8919	0.8553	**TimeDiT (raw)**
width_{50} (diag)	0.05070	0.05901	0.05376	0.05536	0.04454	0.05477	0.05573	0.06286	—
width_{90} (diag)	0.1318	0.1488	0.1353	0.1439	0.1166	0.1348	0.1462	0.1520	—
lower miss_{90} (→0.05)	0.07764	**0.04840**	0.06476	0.05554	0.09595	0.09265	0.05731	0.08331	**TimeDiT (raw)**
upper miss_{90} (→0.05)	0.05945	**0.05591**	0.06616	0.06396	0.1135	0.08966	0.05078	0.06134	**TimeDiT (raw)**
step ($M \times H$)									
RMSE ↓	**0.01244**	0.01245	0.01245	0.01249	0.01295	0.01332	0.01245	0.01253	**CSDI**
CRPS ↓	0.006780	**0.006766**	0.006794	0.006812	0.01272	0.01473	0.006765	0.006874	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4510	0.4811	0.4656	**0.4813**	0.9449	0.9479	0.4849	0.5157	**SBTS**
coverage_{90} (→0.90)	0.8602	**0.8854**	0.8591	0.8782	0.9970	0.9949	0.8869	0.8866	**TimeDiT (raw)**
width_{50} (diag)	0.01321	0.01429	0.01359	0.01434	0.06405	0.07667	0.01434	0.01602	—
width_{90} (diag)	0.03524	0.03794	0.03530	0.03769	0.1609	0.1801	0.03828	0.03968	—
lower miss_{90} (→0.05)	0.07098	**0.05573**	0.07233	0.05890	0.001770	0.002319	0.05676	0.05933	**TimeDiT (raw)**
upper miss_{90} (→0.05)	0.06885	**0.05884**	0.06860	0.06287	0.001221	0.002808	0.05634	0.05408	**TimeDiT (raw)**
rv (M **scalar)**									
MAE ↓	0.01458	**0.01364**	0.01516	0.01420	0.2198	0.2570	0.01381	0.01509	**TimeDiT (raw)**
RMSE ↓ (true, root-last)†	0.01875	**0.01711**	0.01926	0.01779	0.2307	0.2865	0.01727	0.01876	*TimeDiT (raw)*†
CRPS ↓	0.01055	**0.009653**	0.01098	0.01010	0.1913	0.2291	0.009777	0.01168	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4609	**0.4805**	0.4121	0.4805	0	0	0.5059	0.2344	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8594	**0.8945**	0.7812	0.8652	0	0.001953	0.9141	0.5332	**TimeDiT (raw)**
width_{50} (diag)	0.02220	0.02273	0.02027	0.02287	0.06861	0.06689	0.02404	0.01202	—
width_{90} (diag)	0.05413	0.05489	0.04837	0.05458	0.1651	0.1610	0.05753	0.02873	—
lower miss_{90} (→0.05)	0.01758	0.04102	0.05664	**0.05078**	1.000	0.9980	0.02734	0.2891	**SBTS**
upper miss_{90} (→0.05)	0.1230	**0.06445**	0.1621	0.08398	0	0	0.05859	0.1777	**TimeDiT (raw)**

Bank size $N = 16\,384$

Winners (of 18 ranked rows): TimeDiT (raw) 15, CSDI 1, LS4 1, SBTS 1.

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ **trajectory**)									
RMSE ↓	0.04915	**0.04901**	0.04903	0.04996	0.04921	0.05094	0.04923	0.05670	**TimeDiT (raw)**
CRPS ↓	0.02543	**0.02534**	0.02542	0.02610	0.02624	0.02722	0.02535	0.02946	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4604	**0.5103**	0.4755	0.4694	0.4050	0.4664	0.4859	0.4719	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8694	**0.8952**	0.8685	0.8644	0.7905	0.8177	0.8884	0.8553	**TimeDiT (raw)**
width_{50} (diag)	0.05197	0.05936	0.05375	0.05538	0.04454	0.05477	0.05606	0.06286	—
width_{90} (diag)	0.1348	0.1484	0.1357	0.1410	0.1166	0.1348	0.1446	0.1520	—
lower miss_{90} (→0.05)	0.07318	**0.04651**	0.06458	0.06787	0.09595	0.09265	0.05731	0.08331	**TimeDiT (raw)**
upper miss_{90} (→0.05)	**0.05743**	0.05829	0.06689	0.06769	0.1135	0.08966	0.05432	0.06134	**CSDI**
step ($M \times H$)									
RMSE ↓	0.01245	0.01245	**0.01245**	0.01261	0.01295	0.01332	0.01245	0.01253	**LS4**
CRPS ↓	0.006777	**0.006768**	0.006788	0.006930	0.01272	0.01473	0.006761	0.006874	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4493	**0.4818**	0.4648	0.4751	0.9449	0.9479	0.4810	0.5157	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8639	**0.8843**	0.8630	0.8624	0.9970	0.9949	0.8862	0.8866	**TimeDiT (raw)**
width_{50} (diag)	0.01326	0.01431	0.01365	0.01444	0.06405	0.07667	0.01437	0.01602	—
width_{90} (diag)	0.03534	0.03781	0.03544	0.03709	0.1609	0.1801	0.03807	0.03968	—
lower miss_{90} (→0.05)	0.07104	**0.05719**	0.06995	0.06750	0.001770	0.002319	0.05762	0.05933	**TimeDiT (raw)**
upper miss_{90} (→0.05)	0.06506	**0.05847**	0.06702	0.07013	0.001221	0.002808	0.05621	0.05408	**TimeDiT (raw)**
rv (M **scalar**)									
MAE ↓	0.01421	**0.01341**	0.01470	0.01414	0.2198	0.2570	0.01347	0.01509	**TimeDiT (raw)**
RMSE ↓ (true, root-last)†	0.01829	**0.01686**	0.01875	0.01764	0.2307	0.2865	0.01689	0.01876	*TimeDiT (raw)*†
CRPS ↓	0.01028	**0.009484**	0.01066	0.01009	0.1913	0.2291	0.009513	0.01168	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4570	**0.4883**	0.4199	0.4629	0	0	0.4961	0.2344	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8691	**0.8926**	0.7871	0.8574	0	0.001953	0.9141	0.5332	**TimeDiT (raw)**
width_{50} (diag)	0.02189	0.02217	0.01990	0.02194	0.06861	0.06689	0.02286	0.01202	—
width_{90} (diag)	0.05318	0.05322	0.04786	0.05252	0.1651	0.1610	0.05559	0.02873	—
lower miss_{90} (→0.05)	0.01562	0.03906	0.05469	**0.05273**	1.000	0.9980	0.03125	0.2891	**SBTS**
upper miss_{90} (→0.05)	0.1152	**0.06836**	0.1582	0.08984	0	0	0.05469	0.1777	**TimeDiT (raw)**

Bank size $N = 65\,536$

Winners (of 18 ranked rows): TimeDiT (raw) 14, CSDI 2, LS4 2.

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04912	0.04909	**0.04900**	0.05207	0.04921	0.05094	0.04910	0.05670	**LS4**
CRPS ↓	0.02535	**0.02533**	0.02536	0.02768	0.02624	0.02722	0.02524	0.02946	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4670	**0.5046**	0.4717	0.4256	0.4050	0.4664	0.4886	0.4719	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8738	**0.8970**	0.8685	0.8198	0.7905	0.8177	0.8887	0.8553	**TimeDiT (raw)**
width_{50} (diag)	0.05283	0.05886	0.05369	0.05231	0.04454	0.05477	0.05630	0.06286	—
width_{90} (diag)	0.1354	0.1480	0.1353	0.1309	0.1166	0.1348	0.1448	0.1520	—
lower miss_{90} (→0.05)	0.07147	**0.04761**	0.06500	0.08795	0.09595	0.09265	0.05688	0.08331	**TimeDiT (raw)**
upper miss_{90} (→0.05)	**0.05469**	0.05536	0.06647	0.09229	0.1135	0.08966	0.05444	0.06134	**CSDI**
step ($M \times H$)									
RMSE ↓	**0.01244**	0.01245	0.01246	0.01297	0.01295	0.01332	0.01245	0.01253	**CSDI**
CRPS ↓	0.006770	**0.006762**	0.006786	0.007304	0.01272	0.01473	0.006754	0.006874	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4486	**0.4765**	0.4667	0.4406	0.9449	0.9479	0.4820	0.5157	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8639	**0.8810**	0.8627	0.8183	0.9970	0.9949	0.8864	0.8866	**TimeDiT (raw)**
width_{50} (diag)	0.01330	0.01423	0.01367	0.01400	0.06405	0.07667	0.01441	0.01602	—
width_{90} (diag)	0.03542	0.03760	0.03538	0.03433	0.1609	0.1801	0.03807	0.03968	—
lower miss_{90} (→0.05)	0.07001	**0.05884**	0.07025	0.08905	0.001770	0.002319	0.05658	0.05933	**TimeDiT (raw)**
upper miss_{90} (→0.05)	0.06610	**0.06012**	0.06708	0.09265	0.001221	0.002808	0.05701	0.05408	**TimeDiT (raw)**
rv (M scalar)									
MAE ↓	0.01393	**0.01327**	0.01448	0.01432	0.2198	0.2570	0.01314	0.01509	**TimeDiT (raw)**
RMSE ↓ (true, root-last)†	0.01796	**0.01672**	0.01850	0.01815	0.2307	0.2865	0.01657	0.01876	*TimeDiT (raw)*†
CRPS ↓	0.01007	**0.009409**	0.01048	0.01058	0.1913	0.2291	0.009330	0.01168	**TimeDiT (raw)**
coverage_{50} (→0.50)	0.4551	**0.4902**	0.4141	0.4258	0	0	0.4980	0.2344	**TimeDiT (raw)**
coverage_{90} (→0.90)	0.8652	**0.8945**	0.7910	0.8262	0	0.001953	0.9199	0.5332	**TimeDiT (raw)**
width_{50} (diag)	0.02148	0.02197	0.01961	0.02067	0.06861	0.06689	0.02259	0.01202	—
width_{90} (diag)	0.05219	0.05302	0.04666	0.04806	0.1651	0.1610	0.05502	0.02873	—
lower miss_{90} (→0.05)	0.01562	0.03711	**0.05078**	0.06445	1.000	0.9980	0.02930	0.2891	**LS4**
upper miss_{90} (→0.05)	0.1191	**0.06836**	0.1582	0.1094	0	0	0.05078	0.1777	**TimeDiT (raw)**

Bank size $N = 262\,144$

Winners (of 18 ranked rows): TimeDiT (raw) 15, CSDI 2, LS4 1.

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04911	**0.04891**	0.04897	0.05373	0.04921	0.05094	0.04898	0.05670	**TimeDiT (raw)**
CRPS ↓	0.02541	**0.02528**	0.02536	0.02987	0.02624	0.02722	0.02524	0.02946	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4644	**0.5010**	0.4723	0.3552	0.4050	0.4664	0.4904	0.4719	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8691	**0.8972**	0.8696	0.7396	0.7905	0.8177	0.8922	0.8553	**TimeDiT (raw)**
width$_{50}$ (diag)	0.05267	0.05879	0.05349	0.04573	0.04454	0.05477	0.05706	0.06286	—
width$_{90}$ (diag)	0.1344	0.1480	0.1350	0.1137	0.1166	0.1348	0.1460	0.1520	—
lower miss$_{90}$ (→0.05)	0.07501	**0.04675**	0.06653	0.1324	0.09595	0.09265	0.05548	0.08331	**TimeDiT (raw)**
upper miss$_{90}$ (→0.05)	**0.05591**	0.05609	0.06384	0.1280	0.1135	0.08966	0.05231	0.06134	**CSDI**
step ($M \times H$)									
RMSE ↓	**0.01245**	0.01245	0.01245	0.01360	0.01295	0.01332	0.01245	0.01253	**CSDI**
CRPS ↓	0.006771	**0.006760**	0.006784	0.008009	0.01272	0.01473	0.006752	0.006874	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4529	**0.4753**	0.4653	0.3699	0.9449	0.9479	0.4825	0.5157	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8638	**0.8822**	0.8645	0.7321	0.9970	0.9949	0.8849	0.8866	**TimeDiT (raw)**
width$_{50}$ (diag)	0.01332	0.01425	0.01367	0.01230	0.06405	0.07667	0.01444	0.01602	—
width$_{90}$ (diag)	0.03547	0.03759	0.03538	0.02975	0.1609	0.1801	0.03806	0.03968	—
lower miss$_{90}$ (→0.05)	0.07080	**0.05811**	0.06830	0.1332	0.001770	0.002319	0.05829	0.05933	**TimeDiT (raw)**
upper miss$_{90}$ (→0.05)	0.06543	**0.05969**	0.06720	0.1347	0.001221	0.002808	0.05682	0.05408	**TimeDiT (raw)**
rv (M scalar)									
MAE ↓	0.01379	**0.01305**	0.01433	0.01487	0.2198	0.2570	0.01306	0.01509	**TimeDiT (raw)**
RMSE ↓ (true, root-last)†	0.01780	**0.01647**	0.01832	0.01899	0.2307	0.2865	0.01642	0.01876	*TimeDiT (raw)*†
CRPS ↓	0.009987	**0.009273**	0.01038	0.01155	0.1913	0.2291	0.009256	0.01168	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4609	**0.4902**	0.4277	0.3496	0	0	0.4922	0.2344	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8672	**0.8926**	0.8105	0.7500	0	0.001953	0.9180	0.5332	**TimeDiT (raw)**
width$_{50}$ (diag)	0.02140	0.02177	0.01929	0.01770	0.06861	0.06689	0.02231	0.01202	—
width$_{90}$ (diag)	0.05196	0.05261	0.04621	0.04149	0.1651	0.1610	0.05430	0.02873	—
lower miss$_{90}$ (→0.05)	0.01758	0.03906	**0.04688**	0.09766	1.000	0.9980	0.03125	0.2891	**LS4**
upper miss$_{90}$ (→0.05)	0.1152	**0.06836**	0.1426	0.1523	0	0	0.05078	0.1777	**TimeDiT (raw)**

Bank size $N = 1\,000\,000$

Winners (of 18 ranked rows): TimeDiT (raw) 15, LS4 2, CSDI 1.

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04925	**0.04888**	0.04894	0.05523	0.04921	0.05094	0.04911	0.05670	**TimeDiT (raw)**
CRPS ↓	0.02544	**0.02521**	0.02534	0.03179	0.02624	0.02722	0.02525	0.02946	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4624	**0.5072**	0.4688	0.3001	0.4050	0.4664	0.4891	0.4719	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8682	**0.8983**	0.8682	0.6575	0.7905	0.8177	0.8941	0.8553	**TimeDiT (raw)**
width$_{50}$ (diag)	0.05285	0.05901	0.05359	0.04003	0.04454	0.05477	0.05719	0.06286	—
width$_{90}$ (diag)	0.1348	0.1486	0.1349	0.09761	0.1166	0.1348	0.1465	0.1520	—
lower miss$_{90}$ (→0.05)	0.07599	**0.04572**	0.06757	0.1738	0.09595	0.09265	0.05463	0.08331	**TimeDiT (raw)**
upper miss$_{90}$ (→0.05)	**0.05579**	0.05597	0.06427	0.1687	0.1135	0.08966	0.05127	0.06134	**CSDI**
step ($M \times H$)									
RMSE ↓	0.01245	0.01246	**0.01245**	0.01421	0.01295	0.01332	0.01244	0.01253	**LS4**
CRPS ↓	0.006767	**0.006760**	0.006778	0.008688	0.01272	0.01473	0.006750	0.006874	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4532	**0.4761**	0.4672	0.3082	0.9449	0.9479	0.4816	0.5157	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8640	**0.8806**	0.8640	0.6479	0.9970	0.9949	0.8864	0.8866	**TimeDiT (raw)**
width$_{50}$ (diag)	0.01338	0.01428	0.01368	0.01071	0.06405	0.07667	0.01449	0.01602	—
width$_{90}$ (diag)	0.03550	0.03758	0.03538	0.02568	0.1609	0.1801	0.03815	0.03968	—
lower miss$_{90}$ (→0.05)	0.07062	**0.05884**	0.06921	0.1762	0.001770	0.002319	0.05725	0.05933	**TimeDiT (raw)**
upper miss$_{90}$ (→0.05)	0.06543	**0.06061**	0.06683	0.1758	0.001221	0.002808	0.05634	0.05408	**TimeDiT (raw)**
rv (M scalar)									
MAE ↓	0.01373	**0.01295**	0.01426	0.01541	0.2198	0.2570	0.01291	0.01509	**TimeDiT (raw)**
RMSE ↓ (true, root-last)†	0.01772	**0.01637**	0.01819	0.01972	0.2307	0.2865	0.01625	0.01876	*TimeDiT (raw)*†
CRPS ↓	0.009938	**0.009179**	0.01032	0.01241	0.1913	0.2291	0.009143	0.01168	**TimeDiT (raw)**
coverage$_{50}$ (→0.50)	0.4746	**0.4883**	0.4102	0.2930	0	0	0.4727	0.2344	**TimeDiT (raw)**
coverage$_{90}$ (→0.90)	0.8633	**0.8945**	0.7988	0.6641	0	0.001953	0.9238	0.5332	**TimeDiT (raw)**
width$_{50}$ (diag)	0.02115	0.02155	0.01914	0.01493	0.06861	0.06689	0.02217	0.01202	—
width$_{90}$ (diag)	0.05138	0.05213	0.04589	0.03608	0.1651	0.1610	0.05411	0.02873	—
lower miss$_{90}$ (→0.05)	0.01758	0.03125	**0.05078**	0.1328	1.000	0.9980	0.02930	0.2891	**LS4**
upper miss$_{90}$ (→0.05)	0.1191	**0.07422**	0.1504	0.2031	0	0	0.04688	0.1777	**TimeDiT (raw)**